

Distributions introduced by now:

- Discrete: Geometric

Binomial $\rightarrow$ Poisson

- Continuous: Uniform

Exponential (continuous version of Geometric)

Normal : see ~~Binomial~~ Binomial and Poisson!

+ Standardisation:

$$X \mapsto \frac{X - EX}{SD(X)}$$

this will have
zero mean and
variance 1 !

(Open the file Binomial - Poisson - xlsx)

Example with an elevator for 4 persons:

$$X_1, X_2, X_3, X_4 \sim N(75, 12^2) \quad \text{independent random variables}$$

\ / \ /
four persons, entering the elevator.

$$P(X_1 + X_2 + X_3 + X_4 > 320) =$$

(overloading
the lift)

$$= P\left(\frac{X_1 + X_2 + X_3 + X_4 - \overset{4 \cdot 75}{300}}{\sqrt{\underbrace{12^2 + 12^2 + 12^2 + 12^2}_{4 \cdot 12^2}}} > \frac{320 - 300}{24} \right) =$$

$$= 1 - P(\text{st. N.} \leq \frac{20}{24}) =$$

$$= 1 - \Phi\left(\frac{20}{24}\right) \approx 1 - 0.7976 = 0.2024$$

$\approx 20\%$

$$X_i \sim N(75, 12^2)$$

$$X_1 + X_2 + X_3 + X_4 \sim N\left(\underbrace{4 \cdot 75}_{300}, \underbrace{4 \cdot 12^2}_{24^2}\right)$$

(Theorem: the sum of independent Normally distributed random variables is also Normally distributed.)

$$1) E(X+Y) = E(X) + E(Y) \quad (\text{independence not needed!})$$

$$2) \text{Var}(X+Y) = \text{Var}(X-Y) = \text{Var} X + \text{Var} Y$$

if X and Y are independent!

$E(XY) = EX \cdot EY$ because $P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$
and also $f_{X,Y}(x,y) = f_X(x) f_Y(y)$ in the continuous case.